NUNA 370 Townsend St. San Francisco, CA 94107

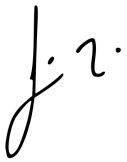
Attention: CMS-0042-NC

Dr. Mehmet Oz
Administrator
Centers for Medicare & Medicaid Services
Department of Health and Human Services
P.O. Box 8013,
Baltimore, MD 21244-8013

Dear Dr. Oz,

Every day at Nuna, we work to create products that make healthcare better for everyone. But we're also patients and caregivers.

These views reflect the personal experiences of Nuna employees, myself included, as we've interacted with the healthcare system, and our own ideas as to where digital health has the greatest opportunity to help.



Jini Kim
Patient, Caregiver, Founder & CEO of Nuna, Inc.

Jini Kim, CEO of Nuna

I serve as the primary caregiver for my brother, my mother, and my father, each presenting unique challenges that demand different approaches and significant time commitments.

My Brother: Navigating Non-Verbal Autism, Epilepsy, and Medication Side Effects

My brother is a mute, Level 3 autistic individual with intellectual and developmental delays, and he experiences grand mal epilepsy. He requires almost **around-the-clock care**. Two of his medications exacerbate his autism-related behavioral issues and negatively impact his sleep quality.

I've observed a **cyclical pattern** in his behavioral issues.

(PC-1, PC-1.a) To better understand and potentially manage these medication side effects, it would be incredibly helpful to have a health management application that helps me systematically track:

- **Mood fluctuations**
- **Dietary intake**
- **Physical activity levels**
- **Sleep quality**
- **Medication timings**

(PC-2.b) Furthermore, connecting this behavioral and lifestyle data with his regular **blood test results** (necessitated by his medications) and his medical records would provide crucial insights into his overall well-being. This integrated approach is vital, especially given his inability to communicate his needs or feelings verbally.

My Mother: Overcoming Language Barriers in Healthcare

My 70-year-old mother's inability to speak English creates significant obstacles in navigating the healthcare system. My father's English proficiency isn't sufficient to advocate for her. While translators are an option, the **slow translation process** consumes valuable appointment time, leaving physicians with less time to address her needs.

Before each appointment, I dedicate about **45 minutes** to gather and translate her health information from Korean to English, then organize it for effective communication with the doctor. During appointments, I'm constantly **translating for my mother** while simultaneously trying to comprehend the doctor's instructions. I record appointments because the intensity of translating often leads me to miss crucial information. Later, I use AI to summarize these recordings.

(PC-1) It would be tremendously beneficial to have a digital solution that could translate for her in real time so her interactions with the health system could be more helpful to her.

Managing medications that require administration 2-3 times a week is particularly challenging, as it's much easier to forget compared to daily, or even every other day, regimens. If she experiences side effects, I send those to the Dr. using the EHR, but by the time we connect with a knowledgeable doctor or nurse (often days later), my mother can't accurately recall the specific side effects she's experienced over the preceding days because there's no system for her to log them as they occur. This entire process is incredibly frustrating.

(PC-1) For someone like my mom who is currently taking and evaluating new medications, it would be valuable to have a tool that helps her keep track of when she needs to take her medications, track side effects as they occur, and get feedback quickly from a physician or pharmacist early in the process.

My Father: Addressing Inaccurate Reporting and Non-Compliance

My father is moderately fluent in English, but his understanding is just poor enough that he doesn't realize what information he's missing or misinterpreting. Consequently, the information he provides to physicians about his health is often **inaccurate**.

Unlike my mother, he frequently exhibits **non-compliance** with care plans. He often misrepresents his exercise habits, medication adherence, and how he's feeling physically (always claiming to feel "great"). This tendency to present himself in a positive light to his doctor, even with inaccurate information, leads to **ineffective care plans**. This inevitably forces me to contact the doctor via the EHR to request revisions. These revised plans are often misinterpreted by support staff who don't know my father, requiring me to **attend subsequent appointments** to ensure the information is accurate and appropriate.

(PC-1) I would greatly benefit from a tool that allows my insights and knowledge as a caregiver to be incorporated into my father's doctor visits **without my physical presence**, reducing the need for multiple appointments. Additionally, a product that **tracks his physician-relevant information** would be invaluable, eliminating his need to relay potentially inaccurate details and mitigating any embarrassment he might feel regarding his non-compliance.

Spouse of Nuna Employee (Anonymous)

I've been a professional care advocate for over 3 decades. I primarily assist families with elderly parents who are experiencing care episodes and work to ensure care continuity and coordinate after discharge. In my experience, there are some key pain points that are potentially addressable with digital health solutions.

The hospital discharge process is poorly coordinated with family members

- The timing of discharge, and the supporting patient discharge criteria is not clearly communicated during an admission, frequently leading to an **unexpected** and **chaotic discharge** process for patients and families and a poor post-acute experience
- Personalized patient discharge instructions are not clearly communicated to patients or caregivers, leaving families with unclear next steps for care.
- The process for assessing the patient's home environment, specifically identifying new patient accessibility needs that require home modifications, is often left to the patient's family with little guidance

(PC-1) It would be helpful to have a digital tool that would electronically communicate discharge information to patient's families, including

- Clinical Discharge criteria and current status vs criteria
- Anticipated timing of discharge
- Personalized, continually updated discharge instructions - based on patients specific post discharge environment (home or Assisted Living or SNF) and patient's clinical history and preferenced

Medication reconciliation is uncoordinated and poorly managed post-discharge

Reconciliation of patients "pre hospitalization" medications with "post discharge" medications often requires scheduling of a post hospitalization PCP visit which can take days to schedule and create a burden to patient and caregivers. It is often difficult to physically get the patient to an appointment merely to adjust medications.

(PC-1) It would be helpful to have clear medication instructions, both post discharge medications and pre discharge medications, as well as remote monitoring of patient after discharge to avoid an unnecessary "face to face" visit or ED visit.

It is difficult to get detailed information about the patient's hospital stay post discharge

After discharge, It is extremely difficult for patients to get information regarding their hospital stay. It's a cumbersome process and "it is like you never existed."

(PC-1) It would be helpful to have a tool that gives patients and care-givers easy access to their hospital records, before discharge, accompanied by an easy-to-understand summary of what occurred during the hospitalization, including clear and concise follow up instructions regarding medications, future clinical needs, therapy notes (while hospitalized, etc).

(PC-1) What apps and data would help you manage the health of your loved ones?

- (Nuna Employee 1) An app that facilitates me and my siblings (and others) participate in the care of my elderly parents. The amount of information is overwhelming and trying to keep distant relatives in sync and addressing their async questions when you are the "primary caregiver" is a lot of stress. Each visit to the doctor, ER, specialist requires vast amounts of communication throughout the "care network".

- (Nuna Employee 2) As a caregiver, it'd be really helpful to have an app that lets me remotely manage the care of my loved ones, including the ability to:
    - Track symptoms and medications
    - Share notes with healthcare providers,
    - Coordinate care with others involved
    - Stay organized with appointments and test results
    - Prepare for appointments and identify questions to ask providers
    - Mood tracking with alerts
    - Aerts in general (for missed meds, etc)